

hold true for the display. On the other hand, 'True-flat' screens do not have this problem.

A flat tube is preferred over a regular curved monitor because of the better geometry offered by the former. True-flat monitors, such as the NEC MultiSync FE791SB and Viewsonic E71f, show no skewed corners when displaying straight lines or grids.

In the 19-inch category, the scene is no different. However, models such as the LG Studioworks 910E don't claim to be flat, and neither is there an additional glass over the picture tube to fool you into thinking it is flat. The NEC MultiSync FP912SB, on the other hand, is an aperture-grille tube-based monitor, and is therefore, a true-flat.

Dot Pitch

Dot pitch plays a significant role in the sharpness of an image. This is the distance between the individual pixels on the monitor screen. A finer dot pitch results in a sharp image, and inclined lines will appear less crooked.

However, aperture-grille tube-based monitors with a tighter dot pitch or line pitch cost a lot more. As of right now, most monitors have diagonal dot pitches ranging from 0.27 to 0.24 mm. If you read that a monitor has a dot pitch smaller than this, make sure it is the diagonal dot pitch, and not the horizontal dot pitch.

Samsung advertises its monitors as having a dot pitch of 0.20 mm, however, this is the horizontal dot pitch, which is naturally smaller than the monitor's diagonal dot pitch value.

The 19-inch NEC MultiSync FP912SB has a fine dot pitch of 0.24 mm, which was the best amongst all the monitors we tested. The Microtek 43F1, a 17-inch monitor, had the coarsest dot pitch: 0.28 mm.

Video Bandwidth

Most of us would love to own monitors that can display 1,280 x 1,024 at 85 Hz, which means more viewing area without headaches due to flicker. This translates to the need for higher video bandwidth, which in turn increases the cost of the monitor. None of the 17-inch monitors we tested, except for the Acer AF715, had such specifications. It offered 200 MHz of video bandwidth, which enables it to reach a maximum resolution of 1,600 x

1,200 at 75 Hz, which is an acceptable figure, even for the most discerning amongst us. However, no other 17-inch monitor came even remotely close to this figure.

In the 19-inch category, the NEC, Philips and Samsung models offered 1,920 x 1,440 resolutions. The NEC offered a refresh rate of 73 Hz, but the others could not offer a flicker-free viewing experience. The NEC could pull this off because of its high video bandwidth of 290 MHz, whereas the rest maxed out at 250 MHz. The Viewsonic E90f+ could only generate 202 MHz of video bandwidth.

Dimensions

By and large, a CRT monitor is the bulkiest device attached to a computer. This is one of the main reasons why LCD monitors have gained such a big foothold in the display market. Vendors have tried to slim down CRTs, but this can only be achieved to a certain extent.

In the 17-inch category, monitor widths ranged from 396 mm to 410 mm, and the depths ranged from 410 mm to 430 mm. A difference of 20 mm is hardly noticeable, so no matter what monitor you choose, be ready to make space for it on your desk. Also, the 19-inch category saw no noticeable variations in dimensions.

Weight Problem

CRTs are heavy, and we are used to this fact. The heaviest part is the picture tube. Unless the technology changes drastically, CRTs will remain heavy. The lightest 17-inch CRT monitor in our test was the Philips 107 E5, weighing in at 13.4 kg. The Viewsonic EF72+SB weighed in at 18.1 kg, and was actually heavier than some of the 19-inch models.

In the 19-inch category, the lightest monitor was LG Studioworks 910E at 17.3 kg. NEC MultiSync FP912SB was the heaviest, at 23.8 kg.

Power Consumption

CRT monitors are power-hungry beasts, and therefore, it's a good idea to power them separately instead of using the cabinet's power supply. A CRT monitor typically needs 70 W of power or

more. This requirement can stress a PC's power supply, which can lead to an unstable system.

Amongst the 17-inchers, the BenqV773 needed 70 W of power, whereas the Acer AF715 demanded a massive 120 W. If an office buys the Acer for its employees, it should be prepared to foot large electricity bills.

In the 19-inch category, the Philips 109 E5 was the most power-efficient. It needed just 65 W of power, which was lower than even the best 17-inch monitor in this test. Even the 109 B6 from Philips needed just 75 W.

These are very impressive figures as far as power consumption is concerned. The biggest power hog in the 19-inch category was—you guessed it—the Acer AC915-130 W.

Extras

All monitors came with a quick start guide, a power cord, and a CD with a manual and a colour profile. The LG Studioworks 700S came with a blue card with a spectral code. This works in conjunction with the Colorific display-tuning utility bundled with the monitor. You can adjust the brightness and contrast perfectly by comparing the colour of the card to the colour on the screen—when they match, your monitor is set perfectly.

Samsung's MagicTune software and Viewsonic's Wizard allow you to tune your monitor's settings to specific applications with minimum hassles.

How They Performed

Both the 17-inch and 19-inch monitors were tested by running 32 selected slides from DisplayMate Video Edition, which pushes the CRTs to the limit. These tests included checking of the geometrical display, sharpness of lines, curves and text, uniformity of the screen, and other problems related to CRT technology—defocusing level shift and ghosting.



Samsung SyncMaster 793MB

What is moiré?

A moiré pattern is a wavy pattern formed on the monitor when very a fine pattern is displayed. These patterns can be due to the graphics board or shadow mask in a CRT monitor when there is cross-talk or noise induced in a signal, or when the electron beam spot size is small. If the spot size is small, it does not adequately fill the hole in the shadow mask—and hence makes the moiré visible.